

# Maximum length of a 3-sum-free set consisting of 2 intervals

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**Theorem 1.** *Let  $A$  be a subset of  $\mathbb{T}$  consisting of 2 disjoint closed intervals with measure at least  $\frac{1}{5}$ . Then,  $A + A - 3 \cdot A = \mathbb{T}$ .*

*Proof.* By monotonicity, it is enough to prove the result in the case  $\mu(A) = \frac{1}{5}$ . Indeed, if  $\mu(A) > \frac{1}{5}$ , then  $A$  contains a subset  $A'$ , still consisting of 2 disjoint closed intervals, with measure exactly  $\frac{1}{5}$ . If  $A' + A' - 3 \cdot A' = \mathbb{T}$ , then, since  $A' \subset A$ , also  $A + A - 3 \cdot A = \mathbb{T}$ . Thus we may assume from now on that  $\mu(A) = \frac{1}{5}$ . We may also assume that both intervals have positive length. Indeed, if one of them has length 0, then the other has length  $\frac{1}{5}$ , and for an interval  $K$  of length  $\frac{1}{5}$  we have  $K + K - 3K = \mathbb{T}$ .

Note that if the statement is true for a set  $A$ , then it is also true for any translate of  $A$  and for the set  $-A$ . Without loss of generality, we can therefore assume  $A = [0, \alpha] \cup [x, x + \beta]$  satisfying the following conditions:

$$0 < \alpha < x < x + \beta < 1, \quad (1)$$

$$\alpha \geq \beta, \quad (2)$$

$$x - \alpha \leq 1 - (x + \beta), \quad \text{i.e.} \quad x \leq \frac{1 + \alpha - \beta}{2}. \quad (3)$$

Indeed, we first choose an interval of maximal length, call its length  $\alpha$ , call the other length  $\beta$ , and translate so that the longer interval starts at 0. We can then write

$$A = [0, \alpha] \cup [x, x + \beta], \quad 0 < \alpha < x < x + \beta < 1, \quad \alpha \geq \beta.$$

Let

$$g_1 = x - \alpha, \quad g_2 = 1 - (x + \beta)$$

be the two gaps between the intervals. If  $g_1 \leq g_2$ , then (3) already holds. If  $g_1 > g_2$ , then we replace  $A$  by the reflected and translated set  $\alpha - A$ . This keeps the longer interval anchored at 0, since

$$\alpha - A = [0, \alpha] \cup [1 + \alpha - x - \beta, 1 + \alpha - x].$$

Writing  $x' = 1 + \alpha - x - \beta$ , the new gaps are

$$x' - \alpha = 1 - (x + \beta) = g_2, \quad 1 - (x' + \beta) = x - \alpha = g_1.$$

Thus the two gaps have been exchanged, and since in this case  $g_2 < g_1$ , the transformed set satisfies

$$x' - \alpha \leq 1 - (x' + \beta).$$

Intuitively, the reflection reverses which side of the longer interval the larger gap lies on; the final translation simply puts that longer interval back in the form  $[0, \alpha]$ . Furthermore, since  $\mu(A) = \frac{1}{5}$ , we have  $\alpha + \beta = \frac{1}{5}$ , and therefore by (2),  $\alpha \geq \frac{1}{10}$  and  $\beta \leq \frac{1}{10}$ . Let us denote  $I = [0, \alpha]$  and  $J = [x, x + \beta]$ . We will prove that  $\mathbb{T}$  is covered by the following 5 intervals:

$$\begin{aligned} R_1 &= I + I - 3 \cdot I = [-3\alpha, 2\alpha], \\ R_2 &= I + J - 3 \cdot I = [-3\alpha + x, \alpha + x + \beta], \\ R_3 &= J + J - 3 \cdot I = [2x - 3\alpha, 2x + 2\beta], \\ R_4 &= I + I - 3 \cdot J = [-3x - 3\beta, -3x + 2\alpha], \\ R_5 &= I + J - 3 \cdot J = [-2x - 3\beta, \alpha + \beta - 2x]. \end{aligned}$$

Note that all of these intervals as written are of the form  $[a, b]$  where  $0 < b - a < 1$ , (we say that the intervals are in *proper* form) and therefore checking if a point  $y \in \mathbb{T}$  is contained in one of these intervals is equivalent to checking if for some integer  $k$ ,  $y + k \in [a, b]$ .

We denote  $U = R_1 \cup R_2 \cup R_3 \cup R_4 \cup R_5$  and  $V = \mathbb{T} \setminus U$ . We assume for the sake of contradiction that  $V$  is non-empty. Identifying  $\mathbb{T}$  with  $[0, 1)$ , note that  $V \subset \mathbb{T} \setminus R_1 = (2\alpha, 1 - 3\alpha)$ , which again is in proper form. The rest of the proof consists in iteratively shrinking this interval through a series of claims on the intervals  $R_i$  until we show that  $V$  is empty. The first claim is the following:

**Claim 1.**  $[2\alpha, \alpha + x + \beta] \subset R_2$ .

*Proof.* We just need to show that  $2\alpha \in R_2$ . In fact, we prove that

$$-3\alpha + x \leq 2\alpha \leq \alpha + x + \beta. \quad (4)$$

The first inequality is equivalent to  $x \leq 5\alpha$ . By assumption (3), we have

$$x \leq \frac{1 + \alpha - \beta}{2} \leq \frac{1 + \alpha - (1/5 - \alpha)}{2} = \alpha + \frac{2}{5}.$$

and since  $\alpha \geq \frac{1}{10}$ , we have  $\alpha + \frac{2}{5} \leq 5\alpha$ . The second inequality is equivalent to  $\alpha \leq x + \beta$ , which is true by assumption (1).  $\square$

We can then conclude that

$$V \subset (\alpha + x + \beta, 1 - 3\alpha). \quad (5)$$

Furthermore, if  $\alpha + x + \beta \geq 1 - 3\alpha$ , then the previous claim implies that  $R_2$  covers the whole interval  $(2\alpha, 1 - 3\alpha)$ , and hence  $V$  is empty, contradicting our assumption. Therefore

$$1 - 3\alpha > \alpha + x + \beta \quad \text{i.e.} \quad x < 1 - 4\alpha - \beta. \quad (6)$$

In particular, the interval in (5) is in proper form. We also note that from inequality (6) we also get the weaker but more convenient

$$x < 1 - (\alpha + \beta) - 3\alpha \leq 1 - 1/5 - 3/10 = 1/2. \quad (7)$$

**Claim 2.**  $[\alpha + \beta + x, 2x + 2\beta] \subset R_3$ .

*Proof.* Similarly to the previous claim, it suffices to show that

$$2x - 3\alpha \leq \alpha + \beta + x \leq 2x + 2\beta. \quad (8)$$

The first inequality is equivalent to  $x \leq 4\alpha + \beta$ . From condition (3) and the fact that  $\alpha + \beta = \frac{1}{5}$ , we have

$$x \leq \frac{1 + \alpha - \beta}{2} = \frac{\alpha - \beta + 5(\alpha + \beta)}{2} = 3\alpha + 2\beta. \quad (9)$$

But since  $\alpha \geq \beta$ , we have  $3\alpha + 2\beta \leq 4\alpha + \beta$ , and we are done.

The second inequality is equivalent to  $\alpha \leq x + \beta$ , which is true by assumption (1).  $\square$

The conclusion of this is that in fact

$$V \subset (2x + 2\beta, 1 - 3\alpha). \quad (10)$$

Similarly as before, if  $2x + 2\beta \geq 1 - 3\alpha$ , then the previous claim implies that  $R_3$  covers the whole interval  $(\alpha + x + \beta, 1 - 3\alpha)$ , and hence  $V$  is empty, contradicting our assumption. Therefore

$$1 - 3\alpha > 2x + 2\beta \quad \text{i.e.} \quad x < \frac{1 - 3\alpha - 2\beta}{2}. \quad (11)$$

and that the interval in (10) is in proper form. Again, we can obtain a weaker but more convenient inequality from (11):

$$x < \frac{1 - 3\alpha - 2\beta}{2} \leq \frac{1 - 2/5 - 1/10}{2} = 1/4. \quad (12)$$

The proof will be completed once we show that  $R_4$  and  $R_5$  cover the interval  $(2x + 2\beta, 1 - 3\alpha)$ . We first show that the union of these two intervals is itself an interval:

**Claim 3.** *The ends of the intervals  $R_4$  and  $R_5$  satisfy the following inequalities:*

$$-3x - 3\beta \leq -2x - 3\beta \leq -3x + 2\alpha \leq \alpha + \beta - 2x.$$

and therefore

$$R_4 \cup R_5 = [-3x - 3\beta, \alpha + \beta - 2x]. \quad (13)$$

Furthermore, this interval is in proper form.

*Proof.* The first inequality is equivalent to  $x \geq 0$ , which is true by assumption (1). The second inequality is equivalent to  $x \leq 2\alpha + 3\beta$ . Now, since  $x \leq 1/4$  by (12), and since  $2\alpha + 3\beta = 2/5 + \beta \geq 2/5 > 1/4$ , this inequality is also true. The third inequality is equivalent to  $\alpha \leq x + \beta$ , which is true by assumption (1). To see that it is in proper form, compute its length directly:

$$(\alpha + \beta - 2x) - (-3x - 3\beta) = x + \alpha + 4\beta.$$

Using (12) and  $\alpha + \beta = \frac{1}{5}$ , we get

$$x + \alpha + 4\beta < \frac{1}{4} + \alpha + 4\beta = \frac{1}{4} + \frac{1}{5} + 3\beta \leq \frac{1}{4} + \frac{1}{5} + \frac{3}{10} = \frac{3}{4} < 1,$$

as required.  $\square$

It remains only to show that the interval in (13) covers the interval in (10):

**Claim 4.**  $R_4 \cup R_5 \supset (2x + 2\beta, 1 - 3\alpha)$ .

*Proof.* We have calculated the ends of both intervals in proper form. Therefore, it suffices to show that

$$-3x - 3\beta \leq (2x + 2\beta) - 1 \leq (1 - 3\alpha) - 1 \leq \alpha + \beta - 2x,$$

where we have shifted both points to check by the integer  $-1$ . We have already established the middle inequality. The first inequality is equivalent to  $5(x + \beta) \geq 1$ , which is true because  $x + \beta \geq \alpha + \beta = 1/5$ . The last inequality is equivalent to  $2x \leq 4\alpha + \beta = 3\alpha + 1/5$ . Since  $x \leq 1/4$  by (12), and we have  $\alpha \geq 1/10$ , we get

$$2x \leq 1/2 = 3/10 + 1/5 \leq 3\alpha + 1/5,$$

and we are done.  $\square$

We have therefore shown that  $V$  is empty, so the proof is complete.  $\square$